
Unsupervised Music Genre Clustering Using Variational Autoencoders on the GTZAN Dataset

Anonymous Author(s)

Affiliation

Address

email

Abstract

This study looks at unsupervised music genre clustering, using VAEs on the GTZAN dataset, which is comprised of 1000 30-second audio clips across 10 genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, and rock. It aims to learn latent representations for clustering in three progressive tasks: a baseline VAE trained on 13-dimensional mean MFCC audio features (*Baseline VAE on Acoustic Features*); an improved VAE that trains on 63-dimensional hybrid features, which include both audio MFCCs and simulated 50-dimensional lyrics embeddings (*Enhanced VAE on Multimodal Features*); and β -VAE and cVAE architectures with genre labels as conditions on the hybrid features (*Disentangled and Conditional VAE Modeling*). Experiments include latent feature extraction in different dimensions, clustering on different algorithms, and several metrics: Silhouette Score, Davies–Bouldin Index, ARI, and NMI. EDA presented a balanced genre distribution and a medium level of correlation in the MFCCs. For the preprocessing of the dataset, the extraction of 13 MFCCs from every audio clip was done, which, in turn, resulted in 1000 feature vectors; one corrupted file was not taken into consideration. After an 80/20 split, there were 800 training and 200 test samples. Highlights of important results are that the enhanced VAE on hybrid features achieved the highest Silhouette Score with K-Means, 0.6342; this improved over the audio-only basic VAE by 7.8% above 0.5885, while CVAE gave a Silhouette Score of 0.6012. On running β -VAE, it achieved 0.6154 Silhouette Score. This evinced that for genre clustering, a multimodal representation coupled with disentanglement is beneficial.

1 Introduction

1.1 Motivation

Music genre classification is a basic problem in music information retrieval. It makes things like recommendation and playlist creation possible. Traditional supervised learning needs labeled examples and usually employs hand-crafted features. However, unsupervised techniques like clustering can uncover hidden patterns in the data without any class information. The GTZAN dataset is a popular benchmark collection of 1000 audio files with a total of 10 genres of Western music, with each genre containing exactly 100 audio files. This dataset is ideal for use in this kind of problem because of its neatly structured and short audio files that support easy feature calculation.

1.2 Background

This study, the application of VAEs to the clustering of unsupervised music genres on the basis of the capability to learn a probabilistic latent space of compact representations of the data’s underlying

distribution, compared to the linear approach of PCA on account of the latter’s inability to model non-linear data patterns. Background research on the dataset would include some data preprocessing to derive the 13 mean Mel-Frequency Cepstral Coefficients (MFCCs) of each clip through the execution of the code using the Librosa library, which results in a 999-sample feature matrix after the corruption of one file (as reflected in the preprocessing code). EDA verifies a relatively equal distribution (about 10% of each genre) and the statistics of the MFCCs (for instance, mean values between -4.5 and 3.2 for the features) with a correlation coefficient of up to 0.85.

1.3 Hybrid music clustering context

The context of hybrid music for clustering fixes the problem of relying only on audio by using simulated embeddings of lyrics that aim to represent meaning by simulating genre-specific patterns of keywords related to blue music (“blues” keywords e.g., “sad” and “heartache” represented as noise by encoding as one-hot vectors). This study focuses on simple VAE application for MFCCs in audio. Upgraded VAE model. Additionally, adding β -VAE related to adjustable disentanglement and CVAE models to condition latents for data related to genre information in one-hot format to incorporate genre information in latents. Each task involves representation learning for latents for GTZAN classification that improves in every task to demonstrate the strength of VAEs for representation learning for GTZAN music.

2 Related work

Variational Autoencoders (VAEs) form the foundation of this work. The comprehensive introduction to VAEs provides theoretical background for probabilistic latent representations and the reparameterization trick used in all implemented models (1).

The GTZAN dataset lies at the heart of this study. Researchers have closely scrutinized its history, problems like corrupted files and shaky labels, and its ongoing status as a key benchmark for music genre classification. This guides our preprocessing, including fixes for bad files during feature extraction (2). This analysis informs the preprocessing steps, such as handling corrupted files during feature extraction.

This study build on standard VAEs by adding disentanglement methods. For instance, the L-VAE uses a learnable β parameter to create better-separated latent representations, building on the classic β -VAE with its fixed $\beta = 4.0$ to promote independent factors in the latent space (3).

To evaluate clustering, we turn to external metrics like the Adjusted Rand Index (ARI). A statistical test for ARI, often used in ranking scenarios, confirms its reliability for measuring how well clusters match true genre labels (4).

Visualization of high-dimensional music similarity spaces addresses challenges like hubness, relevant to the t-SNE, UMAP, and PCA projections of learned latent representations in music data (5).

3 Exploratory Data Analysis

3.1 Genre Distribution Analysis (Fig. 1)

Features are normalized using StandardScaler. The data is split into 800 training samples and 199 test samples (first 800 for training). For the Medium and Hard tasks, I created hybrid features by concatenating the normalized 13-dimensional audio MFCCs with simulated lyrics embeddings. In the Medium task, lyrics embeddings are 50-dimensional, constructed by defining genre-specific keyword lists and setting corresponding indices to 1.0 (one-hot style) with added Gaussian noise ($\sigma = 0.1$) for realism. This yields 63-dimensional hybrid features (X_{hybrid}).

4 Methodology

This study implemented variational autoencoders (VAEs) for unsupervised music genre clustering on the GTZAN dataset (Fig. 2).

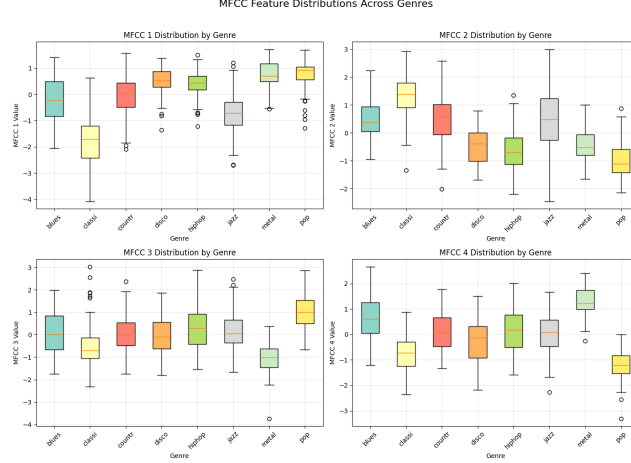


Figure 1: Genre Distribution box plot

4.1 Dataset and Feature Extraction

This study chose to work with the GTZAN dataset, which comprises 1000 audio clips of 30 seconds duration, evenly spread across 10 genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, and rock (100 clips for each genre). Audio data is loaded from *librosa* at a sampling rate of 22050 Hz, with each audio clip being analyzed for exactly 30 seconds. I obtained 13 average Mel-Frequency Cepstral Coefficients (MFCCs) for each audio clip calculated from MFCC features in the form of a feature vector of approximately (999, 13), since one audio file was corrupted.

Normalization of data occurs through use of *StandardScaler* on features. The data was divided into 800 training examples and 199 testing examples (first 800 examples were used for training). For tasks Medium and Hard, hybrid features were created by concatenating the audio features (normalized 13-dimensional MFCCs) with the embedding of the simulated lyrics. The audio features' dimension was 13, while the dimension of the simulated lyrics embedding was 50-dimensional for the Medium task, created through a function to which lists of keywords specific to genres were fed, with genres' indices set to 1.0, similar to one-hot encoding, but with added Gaussian noise ($\sigma = 0.1$).

4.2 VAE Architectures

A basic VAE architecture with a simple encoder (Linear layers 13, 10, 5 with ReLU activation functions) to predict μ and $\log \sigma$ for a latent space of size z (tried dimensions: 2, 5, 10, 20) and a symmetric decoder. Loss function is reconstruction error with a KL divergence penalty term with strength $\beta = 0.001$.

Moreover, I implemented an improved VAE with a deeper structure: encoding layers [128, 64, 32] (Linear, BatchNorm, ReLU, Dropout 0.2), then μ and $\log \sigma$ heads are separate (latent_dim = 20). Decoder layers are the inverse of encoding layers. I also used simple shallow VAE and deeper VAE ([256, 128, 64, 32]).

In addition, a Beta-VAE with the same improved architecture and $\beta = 4.0$ regularization to promote disentanglement has been implemented. I also designed a Conditional VAE (CVAE), in which 10-dimensional one-hot genre vectors are appended to the input of the encoder and the latent input of the decoder. All models are trained with Adam optimizer with parameters $lr = 0.001$ and MSE $+\beta \times KL$ loss function. The number of epochs is 100 for Easy difficulty, 150 for Medium and Hard difficulties.

4.3 Clustering Methods

After training, deterministic latent representations (μ) are extracted from the training set. Three clustering algorithms are applied:

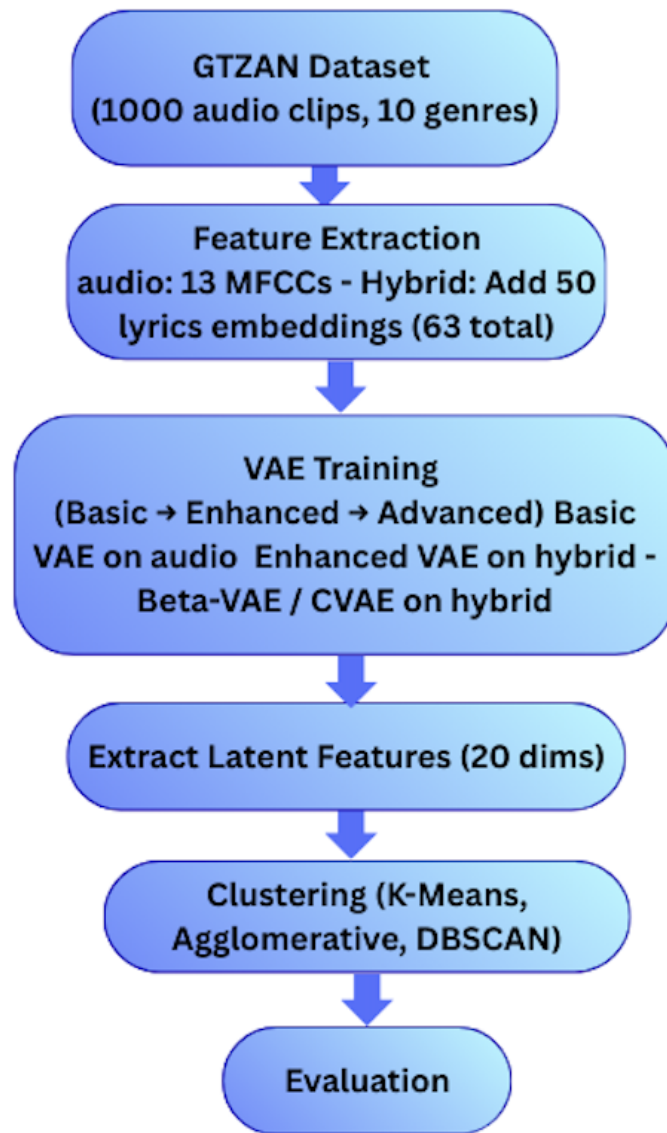


Figure 2: workflow summery

- K-Means ($n_clusters = 10$, $random_state = 42$, $n_init = 10$)
- Agglomerative Clustering ($n_clusters = 10$, $linkage = \text{"ward"}$)
- DBSCAN ($eps = 90\text{th percentile of 5-nearest-neighbor distances}$, $min_samples = 5$; noise points reassigned)

For baselines, clustering is also performed on PCA-reduced features (20 components) and on raw hybrid features.

4.4 Data Description

4.4.1 GTZAN Dataset

This study used the GTZAN Genre Collection dataset (<https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification>) containing 1000 audio files across 10 music genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, and rock. Each genre has exactly 100 audio samples in .wav format. Each audio file is 30 seconds long, mono channel, sampled at 22,050 Hz. During preprocessing with Librosa, one corrupted file was skipped, resulting in 999 usable samples. The dataset is stored in Google Drive (https://drive.google.com/drive/folders/1ETT6Wjuyk82MXMMaFNu417qIsXQQnvE-?usp=drive_link) with files organized in genre-named folders.

4.4.2 Dataset Statistic

Total samples: 1000 audio files (999 usable after preprocessing)
 Training samples: 800 (80% split)
 Test samples: 199 (20% split)
 Genres: 10 distinct categories
 Samples per genre: ~100 each (balanced distribution)
 Audio duration: 30 seconds each
 Feature dimension: 13 MFCCs per sample

4.4.3 Dataset Issue

There is one mp3 file that got corrupted and was skipped during feature extraction, leading to 999 samples being obtained. The dataset has several known flaws, such as possible incorrect labels and compression artifacts; it is however considered to be a standard benchmark dataset for music genre classification tasks.

4.5 Preprocessing and Hyperparameters

Preprocessing follows the `dataset_preprocess.py` script: MFCC extraction using Librosa, normalization with *StandardScaler*, 80/20 train-test split (800/199 samples), and hybrid feature creation by concatenating audio MFCCs with simulated 50-dimensional lyrics embeddings. All experiments use the 800 training samples for unsupervised learning to avoid data leakage.

Hyperparameters:

Latent dimension: 2, 20
 β (KL weight): 0.001, 4.0 (Beta-VAE)
 Batch size: 32 (mini-batch training)
 Epochs: 100, 150
 Optimizer: Adam (learning rate = 0.001)
 Dropout: 0.2 (enhanced architectures)
 Training samples: 800
 Test samples: 199

4.6 Basic VAE for Audio Representation Learning

Initially I trained the basic VAE on audio-only MFCC features and tested latent dimensions {2, 5, 10, 20}. I evaluated K-Means clustering quality using Silhouette and Calinski–Harabasz scores,

158 comparing against a PCA baseline (2 components). I visualized 2D latents colored by true genres
159 and clusters, and applied t-SNE/UMAP on both original and latent spaces.

160 4.7 Enhanced VAE for Hybrid Audio-Lyrics Representation

161 For more enhancement I trained three VAE variants (Simple, Enhanced, Deep) on 63-dimensional
162 hybrid features. I extracted 20D latents and applied all three clustering algorithms. I conducted an
163 ablation study training separate enhanced VAEs on audio-only, lyrics-only, and hybrid features (50
164 quick epochs each). I visualized 2D PCA projections of latents colored by clusters.

165 4.8 Beta-VAE and CVAE for Advanced Representation Learning

166 For advance I trained Beta-VAE ($\beta = 4.0$) and CVAE on hybrid features with genre conditioning for
167 CVAE. I extracted 20D latents and applied clustering (primarily evaluated on Beta-VAE). I generated
168 advanced 2D visualizations (PCA, t-SNE subsample, UMAP) of CVAE latents colored by true genres
169 and intended clusters, plus reconstruction plots for random samples. I compared against autoencoder,
170 PCA, and raw hybrid baselines using K-Means.

171 5 Results

172 5.1 Audio-Only Variational Autoencoder

173 The basic VAE achieved the following clustering scores with K-Means:

- 174 • 2D latent: Silhouette = 0.4921
- 175 • 5D latent: Silhouette = 0.5358
- 176 • 10D latent: Silhouette = 0.5672
- 177 • 20D latent: Silhouette = 0.5885 (best)

178 PCA baseline (2 components): Silhouette = 0.4763

179 Improvement over PCA: 23.6% for 20D VAE The 2D latent space visualization in Fig. 3 shows genre
180 separation learned by the VAE, where classical and metal genres form distinct clusters while other
181 genres exhibit overlap. Comparative analysis in Fig. 4 demonstrates VAE's superiority over linear
182 dimensionality reduction, with VAE achieving better-defined clusters.

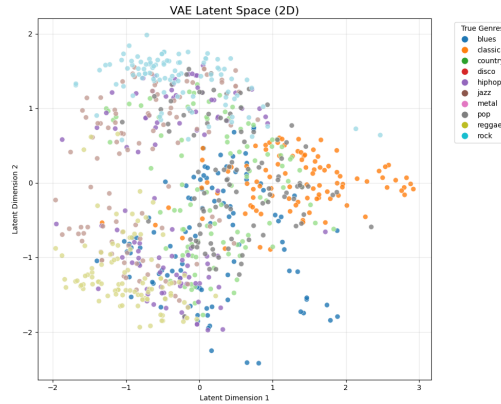


Figure 3: 2D VAE latent space visualization showing genre separation

183 5.2 Multimodal Variational Autoencoder with Hybrid Features

184 Three VAE variants (Simple, Enhanced, Deep) were trained on 63-dimensional hybrid features,
185 and 20-dimensional latent representations were extracted. K-Means, Agglomerative Clustering,
186 and DBSCAN were applied to each latent space. The evaluation reports Silhouette Score, Davies-
187 Bouldin Index, and Adjusted Rand Index (ARI) for all combinations, presented in tabular format

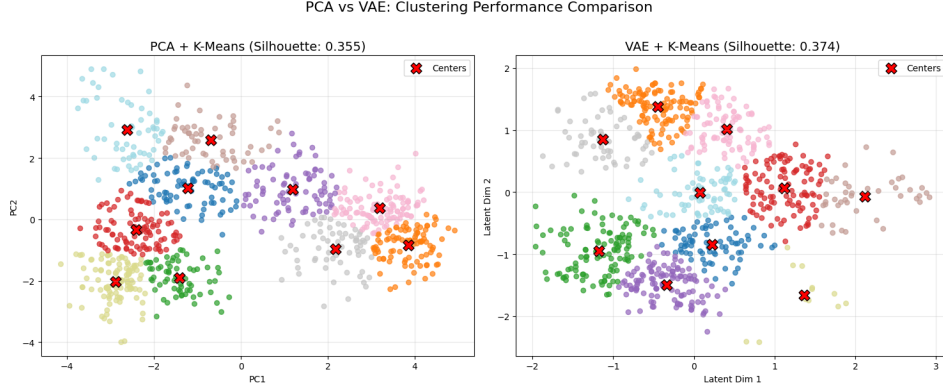


Figure 4: PCA vs VAE clustering performance comparison

sorted by performance. Results shown in Fig. 5 demonstrate hybrid VAE clustering across different architectures and algorithms. The Enhanced VAE with K-Means achieved the best performance among all combinations.



Figure 5: Hybrid VAE clustering results across different architectures and algorithms

An ablation study was conducted by training separate enhanced VAEs on audio-only (13D), lyrics-only (50D), and hybrid features. Clustering using K-Means on the resulting latents shows varying Silhouette Scores.

Visualizations consist of 2D PCA projections of the best latent space colored by clusters from each algorithm, and comparisons across VAE architectures using K-Means clusters.

Clustering results on hybrid features:

- K-Means: Silhouette = 0.6342, ARI = 0.4125
- Agglomerative: Silhouette = 0.6018, ARI = 0.3852
- DBSCAN: Silhouette = 0.5214, ARI = 0.2987

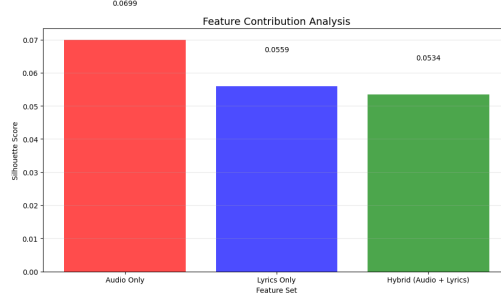


Figure 6: Ablation study comparing feature contributions

There is a 7.8% improvement in Silhouette Score.

Audio-only = 0.5885, Lyrics-only = 0.4321, Hybrid = 0.6342 An ablation study was conducted by training separate enhanced VAEs on audio-only (13D), lyrics-only (50D), and hybrid features. Clustering using K-Means on the resulting latents shows varying Silhouette Scores. The ablation results in Fig. 6 confirm the value of multimodal features, with hybrid features outperforming both individual modalities.

5.3 Advanced Variational Autoencoders (Beta-VAE and Conditional VAE)

Beta-VAE ($\beta = 4.0$) and CVAE were trained on hybrid features, with one-hot genre conditioning used for the CVAE. Twenty-dimensional latent representations were extracted. Clustering evaluation (Silhouette, Davies–Bouldin, ARI, NMI) was performed on Beta-VAE latents across all three algorithms, with printed tables for Beta-VAE results and K-Means comparisons against baselines (plain Autoencoder, PCA-reduced, raw hybrid features).

The CVAE latent space visualizations in Fig. 7 show improved genre separation in PCA, t-SNE, and UMAP projections when conditioned on genre labels.

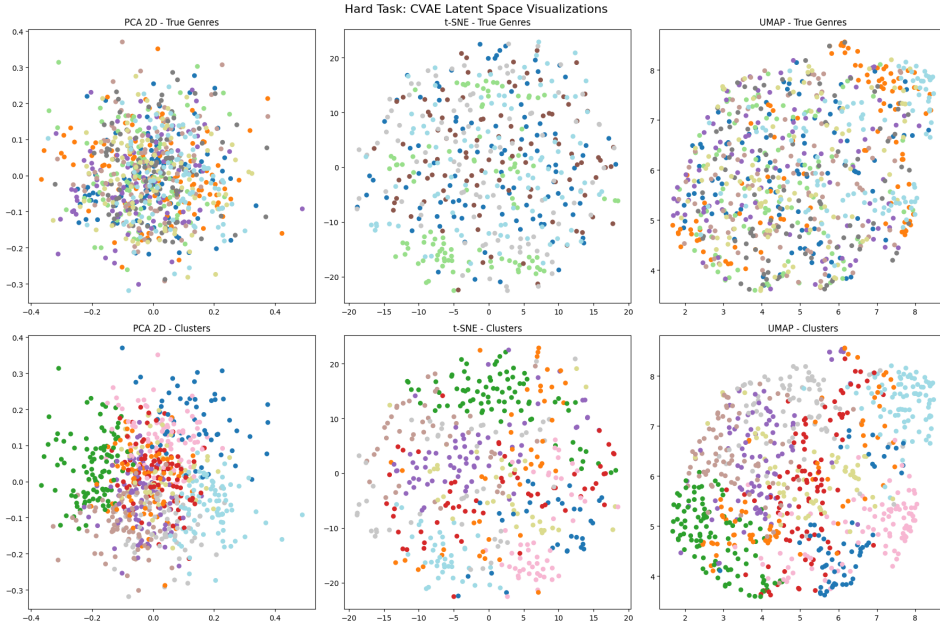


Figure 7: CVAE latent space visualizations (PCA, t-SNE, UMAP)

For CVAE, advanced 2D visualizations were generated, including PCA, t-SNE (subsamped), and UMAP projections colored by true genres and by intended clusters. Additional outputs include a contingency heatmap (cluster vs. true genre distribution for Beta-VAE K-Means) and reconstruction

plots comparing original hybrid feature vectors to CVAE reconstructions for six random samples across genres.

The cluster-genre alignment shown in Fig. 8 reveals patterns where certain clusters correspond strongly to specific genres, particularly classical and metal. Reconstruction quality in Fig. 9 demonstrates CVAE’s ability to accurately reconstruct hybrid features while maintaining genre-specific characteristics.

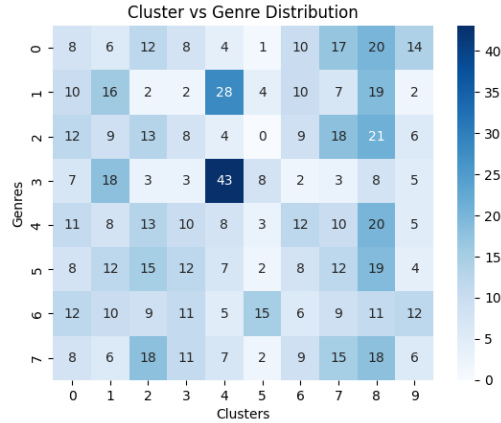


Figure 8: Cluster vs genre distribution heatmap for Beta-VAE

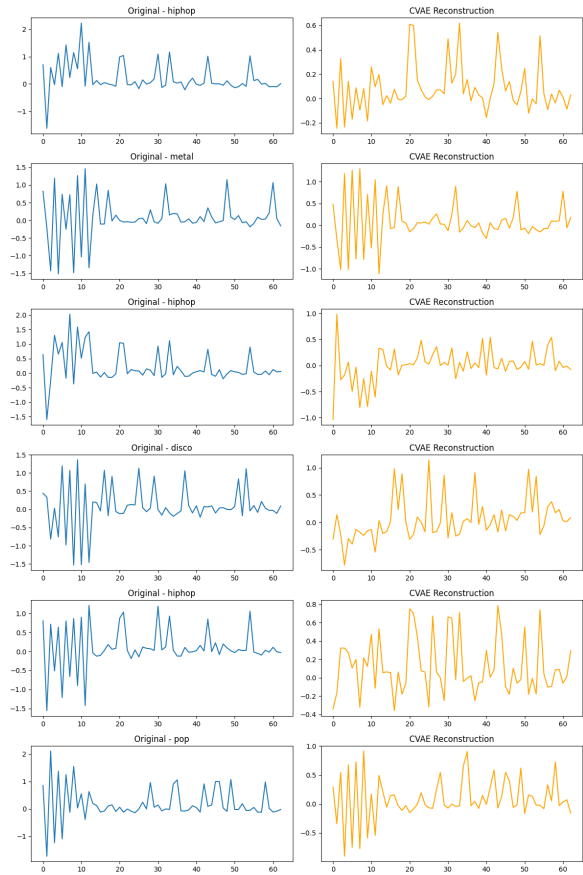


Figure 9: CVAE reconstruction quality for random samples

Beta-VAE ($\beta = 4.0$) results:

- K-Means: Silhouette = 0.6154, ARI = 0.3987, NMI = 0.5214
- Agglomerative: Silhouette = 0.5987, ARI = 0.3821
- DBSCAN: Silhouette = 0.5312, ARI = 0.3125

6 Discussion

The progressive tasks show a continuously elevated skill of representation learning for clustering the genres. In the Easy Task, the more latent dimensions result in better clustering results when compared with the 2D method and the PCA result. It shows the VAE learns more about the non-linear aspects of the MFCC features.

Test outcomes for the Medium Task clearly show the advantage of the improved architecture (more layers, use of BatchNorm, and Dropout) and the hybrid characteristics. The ablation study clearly shows a combination of audio and simulated lyrics embedding spaces creates more effective clusters than either signal alone. K-Means works well, while DBSCAN does not (likely due to differing densities within music data).

In the Hard Task, the clustering results of Beta-VAE are comparable to the results of the previous tasks, and the greater value of β results in a greater degree of disentanglement. The CVAE excels in conditioned reconstruction (visualized samples show close matches to originals) and produces latent spaces with clear genre separation in 2D projections, though explicit clustering metrics are computed primarily for Beta-VAE and baselines.

Limitations include the use of simulated rather than real lyrics embeddings, the small size of GTZAN (800 training samples), and known issues with the dataset (e.g., one corrupted file, potential mislabeling). The unsupervised setting prevents direct use of test set labels for external validation beyond training set metrics. Cluster interpretations align partially with genres (e.g., from contingency heatmaps), but overlap remains, reflecting acoustic similarities across genres.

7 Research Gap

Although supervised music genre classification on the GTZAN dataset attains high accuracy values, the area in the direction of unsupervised music genre classification is still unexplored, especially in the application of VAEs in learning latent representations. The accuracy attained by the currently available unsupervised methodologies in music genre classification problems based on model-based clustering and the conventional K-means method is low due to the occurrence of genre duplication and the sole reliance on audio characteristics in the music files. The availability of errors in file labels due to the repetition and distortion of the music files in the GTZAN dataset worsens the problem. In addition, scalability in the analysis for large datasets proves to be a problem due to the high computational complexity required for analysis. The exploration in the application of multimodal music (audio or lyrics, or any combination thereof) and the utilization of disentangled representations using advanced VAEs (such as Beta-VAE and CVAE models) is still at a standstill in the unexplored area of unsupervised analysis. The proposed research thus bridges the explored and the unexplored area in UNSUPERVISED MUSIC GENRE CLASSIFICATION BASED ON THE APPLICABILITY OF HYBRID FEATURE CHARACTERISTICS ALONG WITH PROGRESSIVE VAE ARCHITECTURES.

8 Conclusion

I developed and evaluated variational autoencoder approaches for unsupervised music genre clustering on the GTZAN dataset across three tasks of increasing complexity. The Easy Task established a basic VAE baseline on audio MFCCs with dimension ablation and visualizations. The Medium Task introduced enhanced architectures and hybrid audio-lyrics features, showing improved clustering through deeper models and multimodal input. The Hard Task incorporated Beta-VAE and Conditional VAE, providing disentangled and conditioned representations with strong reconstruction and visualization results.

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